



Full length article

Life cycle impact on climate change caused by metal production from deep sea manganese nodules versus land-based deposits

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ABSTRACT

At the "UN Climate Change Conference" (COP27), France called for a ban on deep sea mining. The topic has gained momentum over the last years as some businesspeople and politicians are promoting the mining of manganese nodules as a less CO₂-intensive way to meet the rising demand for the battery metals nickel, copper, cobalt and manganese. In this study, we modelled the impact on climate change caused by metal production from manganese nodules vs. mining on land. The total climate impact of producing these metals from manganese nodules (~2.0 kg CO₂eq/kg dry nodules) compared to current land-based projects is about 27% lower. Comparing it to values reported in the literature by industry associations and by the ecoinvent database, the climate impact from manganese nodules is about 16% lower or 28% higher, respectively. Based on their monetary values, the allocation of the total climate impact to each metal assigns the highest contribution to FeMn. Our research shows that producing Ni, Cu, Co and FeMn from manganese nodules could lead to higher or lower impact on climate change depending on the model input data and methodological choices. The authors share the opinion that climate change should not be the central argument regarding the realization of manganese nodule projects. The discussion should instead explicitly include further environmental impacts, e.g., biodiversity or plumes as well as economic considerations.

1. Introduction

In recent years, polymetallic manganese nodules from the sea floor of the Pacific Ocean's Clarion Clipperton Zone (CCZ), which covers an area of about five million km², have been the subject of a controversial debate on future sources for cobalt, nickel, manganese and copper (European Union, 2020; Hein et al., 2020; Heinrich et al., 2020; World Bank Group, 2017). With average grades of 0.21% cobalt, 1.3% nickel, 1.1% copper and 25% manganese per dry weight (approximately 21% water content, dried at relatively low temperatures) as estimated by Dames & Moore (1996), these nodules appear to be an attractive source of metals for strategic technologies and sectors in the EU (Bobba et al., 2020). Apart from questions of technological and economic feasibility, as well as licences and permits, environmental risks have been at the centre of a controversial international debate about nodule mining. For example, impact on biodiversity as a result of disturbing the sea floor and generating sediment plumes are subject to intensive investigation

(Gillard et al., 2019; Gollner et al., 2017; Gollner et al., 2022; Niner et al., 2018). This risk of irreversible environmental damage represents one of the major reasons why various major companies have voted for a moratorium on deep sea mining (WWF, 2022), and why Germany, among other countries, has advocated for a precautionary pause in November 2022 (BMUV, 2022). However, industrial start-ups present the potential reduction in impact on global warming as a powerful argument in favour of manganese nodule exploitation, portraying its environmental impact as a necessary evil to address the climate crisis (Impossible Mining Inc., 2022; Jaekel, 2022; The Metals Company, 2022c). Indeed, greenhouse gas (GHG) emissions have become a key indicator of sustainable production, and an increasing number of companies, also in the mining industry, have set evidence-based reduction targets (Bjørn et al., 2021). However, scientific studies investigating the hypothesis that metal production from manganese nodules are less GHG-intensive compared to conventional sources are scarce. In fact, until today, only two studies have performed a comparative life cycle

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assessment (LCA) on this topic. Since different goals and scopes highly affect an LCA study's outcome, we describe the most important assumptions of these two studies in the following and elaborate the research gap.

The first study from Paulikas et al. (2020), which was also promoted by the industrial start-up The Metals Company (2022a), found that metal production from manganese nodule processing potentially causes significantly lower CO₂-eq than from land ores. In their research on climate impact, however, they chose as a functional unit a demand for one billion electric vehicles until 2047, resulting in a study design that only processes 5% of the manganese in the nodules into a marketable end product. We consider this design to be a major limitation to deriving valid findings for the hypothesis that metal production from manganese nodules is less GHG-intensive. It seems reasonable instead to follow an approach investigating a zero-waste value chain (Friedmann et al., 2017; Keber et al., 2020; Paulikas et al., 2020; Sommerfeld et al., 2018). This would allow analysis of the processing of all contained Ni, Cu, Co and, in particular, FeMn (manganese) into marketable products within product system boundaries. This hypothesis is based on published data from the few commercial projects proposing this approach. Volkmann et al. (2019) consider a zero-waste value chain to be a crucial condition for making manganese nodule production economically viable. Furthermore, Paulikas et al. (2020) argue that, with the future application of deep sea mining and flexibility in choosing onshore locations, electricity demand for onshore operations would be fully provided by hydropower. This assumption is countered by a pre-feasibility study of a commercial-scale plant in India (The Metals Company, 2022b), where hydropower's share in total electricity generation will remain below 20% in the future (IEA, 2023). Even in countries with a high contribution of hydropower to electricity generation, a plant would compete with other electricity users and presumably lead to displacement effects, without forcing the expansion of low-carbon electricity sources. Operating a plant with other low-carbon electricity sources (as also recommended by The Metals Company (2022b) for the plant in India) seems possible, but applies equally to land-based mines. Therefore, it seems advisable to calculate two scenarios: a conservative scenario including current power mixes for the refining of manganese nodules and land ores, and another scenario including low-carbon electricity sources.

The second study by Alvarenga et al. (2022) addresses the limitations in Paulikas et al. (2020). Their research concludes that the product basket of metals sourced from manganese nodules could result in a 16–38% reduction in climate impact compared to their land-based counterparts, assuming a novel hydrometallurgical refining process for the manganese nodules. On the other hand, a process route based on conventional pyrometallurgical smelting and reduction processes for recovering FeMn, followed by hydrometallurgical processing to recover all other metals of economic interest (Ni, Co, Cu), has been discussed most recently as an application in manganese nodule processing. Moreover, Alvarenga et al. (2022) determined the climate impact of comparative terrestrial mines by using data provided by metal commodities associations (for Ni, Cu, Co) and a dataset (for FeMn). However, they also mention that the comparative data is difficult to analyze in more detail, e.g. by comparing hot spots and performing sensitivity analysis, since the datasets are only available in aggregated format. A higher disaggregation of the process models from land-based metal production would allow more transparency in methodological assumptions, such as system boundaries and allocation and a deeper understanding of differences in results.

Our study addresses these limitations and aims to complement published studies to shed further light on the question whether manganese nodules are a source for less GHG emission intensive metal production. We integrate (1) a zero-waste approach for nodule processing (Sharma, 2017; Sommerfeld et al., 2018), (2) pyro-hydrometallurgical route (Sharma, 2017; Sommerfeld et al., 2018; The Metals Company, 2022a) and (3) further comparative data for terrestrial mines. Additionally, we (4) calculate an alternative scenario

for the power mix composition for the manganese nodules and the land-based models.

2. Material and methods

The objective of this study is to analyze the impact on climate change caused by the production of nickel (Ni), copper (Cu), cobalt (Co) and ferromanganese (FeMn) from manganese nodules from the Clarion Clipperton Zone (CCZ) and compare it to existing land-based equivalents.

We calculated the climate impact based on the International Organization for Standards (ISO) Life Cycle Assessment (LCA) guidelines 14,040 (DIN EN ISO 14,040, 2020) and 14,044 (DIN EN ISO 14,044, 2020), and addressed climate change as the only impact category. We compared two product systems: (1) the recovery of these metals from manganese nodules extracted by deep sea mining, and (2) the recovery of an equivalent metal product basket (see Alvarenga et al. (2022)) from a compilation of existing land sources. The functional unit or more precisely the declared unit (European Commission, 2010) was set to the supply of 15.3 g of Ni, 11.7 g of Cu, 2.2 g of Co and 338 g of FeMn, which in total equals the yield of valuable metals from processing 1 kg of dry manganese nodules (wet mass: 1.66 kg).

For impact assessment, we used the GWP₁₀₀ characterization factors from IPCC (2013). Since nickel-equivalents (Ni-eq) are a common indicator in deep sea mining literature, we provide our results in CO₂-eq and Ni-eq (Sharma, 2017). For the sake of quality control, all inventories, assumptions and results were reviewed by a panel with strong expertise in life cycle assessment, mining and metallurgy. In addition, we compared our results for climate change impact from the deep sea LCA model with literature data for terrestrial metal production.

2.1. Description of the manganese nodules system and inventory

The product system for the production of metals from manganese nodules includes all processes from cradle to gate. For the foreground system, we collected data from literature and feasibility studies (Dames & Moore and EIC Corporation, 1977; Heinrich et al., 2020; Kirchain et al., 2019). There is no industrial manganese nodule mining taking place yet, which is why there are no field-proven data. However, the basic metallurgical processes, namely a pyrometallurgical extraction of ferromanganese described by Dames & Moore and EIC Corporation (1977), are still discussed today. Electricity, all material inputs such as gypsum, chemicals, silica sand, oxygen and hydrogen as well as waste treatment activities represent the background system for which we used the ecoinvent v.3.7.1 cut-off database. Capital goods and infrastructure were excluded from the scope unless the flows have been listed in ecoinvent.

As mining of manganese nodules has not yet been performed on an industrial scale, in consultation with industry experts, we selected a plausible value chain for the deep sea process route. The first step is the collection of manganese nodules from the seafloor. This is done using collectors that operate on the deep seafloor and are remotely controlled by a surface ship. The nodule collectors are each connected to a hose and a riser that pump the manganese nodules to the sea surface (ISA, 2021). We calculated the energy needed for the collectors, pumps and a mining vessel according to data from Kirchain et al. (2019), with diesel fuel as the primary energy source used either in the vessel's generator set or directly in a diesel-powered water pump. For the energy consumption of the vessel, data was available in joules, while the ecoinvent 3.7.1 (2021) dataset that best represents marine propulsion is specified in ton-kilometers (tkm). To represent energy consumption in tkm, we used estimates on heavy fuel oil consumption per tkm according to Notten and The Green House (2021). Activities in the life cycle for the vessel and the water pump, according to background datasets from ecoinvent 3.7.1 (2021), include, e.g., vessel maintenance and the pump construction. In contrast, for the collector we modelled the energy used

solely as diesel burned in the vessel's generator set. The exact numbers used are provided in Fig. 1. All tons in the study are metric tons.

The second step is the transport of the manganese nodules from the offshore collection site to the on-shore pyro- and hydrometallurgical refining plant. We assume the processing plant would be located at the closest port to the Clarion Clipperton Zone (CCZ), in Lázaro Cárdenas, Mexico, which is about 1040 nautical miles (approx. 1930 km) away from the exploitation site.

Finally, the manganese nodules are processed and refined onshore into the final products cathode Ni, Co powder, cathode Cu and FeMn. For the copper-sulfide (CuS) and zinc-sulfide (ZnS) products, no impact were allocated because in Dames & Moore and EIC Corporation (1977) there is no further information about these products and the production volumes are very low compared to the other metals. A sensitivity analysis that assumes a high market price for CuS and ZnS of 3 €/kg decreases climate impact for each metal by less than 1%. After grinding and drying of the manganese nodules, a Ni-Cu-Co rich flow is separated from a FeMn rich flow by coke reduction. Reoxidizing and sulfurizing the Ni-Cu-Co rich part forms a matte. Using sulfuric acid, further impurities are removed, and a Ni-Cu-Co-bearing solution is formed. From this solution, Ni and Cu are separated by a series of hydrometallurgical purification steps. Cathode Cu and Ni are produced from the acidic solution by electrowinning. Finally, the Co is removed from the solution by precipitation with hydrogen sulfide. For a more detailed description of the process, please see supplementary material section I and Haynes et al. (1983).

We modelled the LCI using the in- and output tables from Dames & Moore and EIC Corporation (1977). This study also provides our reference process in terms of total production, with the total amount of dry manganese nodules from the previous process steps of collection and transport of the manganese nodules scaled proportionally to 1120 kt wet nodules (674 kt dry mass) collected annually. According to Dames & Moore and EIC Corporation (1977), the moisture content of the reclaimed wet manganese nodules is 40%. In the output list for Ni, two products are given, cathode Ni (10 kt) and Ni powder (0.18 kt). We summed them up without any weight adjustments. We modelled the upstream production only of typical fossil resources needed like coal, coke and oil, without considering them for combustion or reduction

purposes, as the study also includes the amount of stacked off-gasses and the proportion of CO₂ in the emissions (Dames & Moore and EIC Corporation, 1977). Validation on the basis of carbon balance showed that this value is rather under- than overestimated (see Section 3.5). Electrodes of the electric furnaces were assumed to consist of 70% graphite and 30% mineral oil (Wang et al., 2001). The mineral oil used for the preparation of electrodes was added to the amount of fuel oil already in the inventory as a fossil resource. As there is no specific dataset for the reagent LIX 64n used in the ion exchange process step, we modelled it as a generic organic chemical. For the flocculants used for waste treatment, we could not find a specific dataset either, so we assumed it to be polyacrylamide. The filter aid was assumed to be silica sand. The inputs needed for the cooling tower and boiler water used to clear and soften plant water, as well as electrowinning additives were neglected due to their minor importance. As gaseous emissions, we only included the final stacked off-gasses. We neglected air emissions that do not contribute to climate change such as oxygen, nitrogen and water vapor and therefore only modelled carbon dioxide.

For ammonia (NH₃) we assume that the amount of 581 kt, given by Dames and Moore and EIC Corporation (1977), is too high because NH₃ is only used for neutralization of the acids, mainly sulfuric acid (H₂SO₄). Stoichiometrically, the neutralization of one gram of H₂SO₄ requires 0.35 g of NH₃, resulting in 8.63 kt of NH₃ per annual production volume. This is a conservative assumption, as lime and limestone are used for neutralization in addition to NH₃.

For modelling the waste effluents (solids and liquids) we used Dames & Moore and EIC Corporation (1977). The 150,600 lbs/hr solid slag was converted to 541 kt per year and modelled with a process for waste Ni smelter slag. The waste gypsum given at 1860 lbs/hr for solids and 5 gpm (gallons per minute) for liquids was converted to 15.7 kt per year, assuming that the density of liquids is equivalent to water. This flow was modelled using a waste gypsum dataset. Finally, all the remaining waste flows (solids and liquids) were added up to 446 kt and processed for landfill inert waste. A detailed process flow diagram including the flow amounts used in our model can be found in Fig. 1. Additionally, supplementary material table S 1 provides all the background processes used to model the flows.

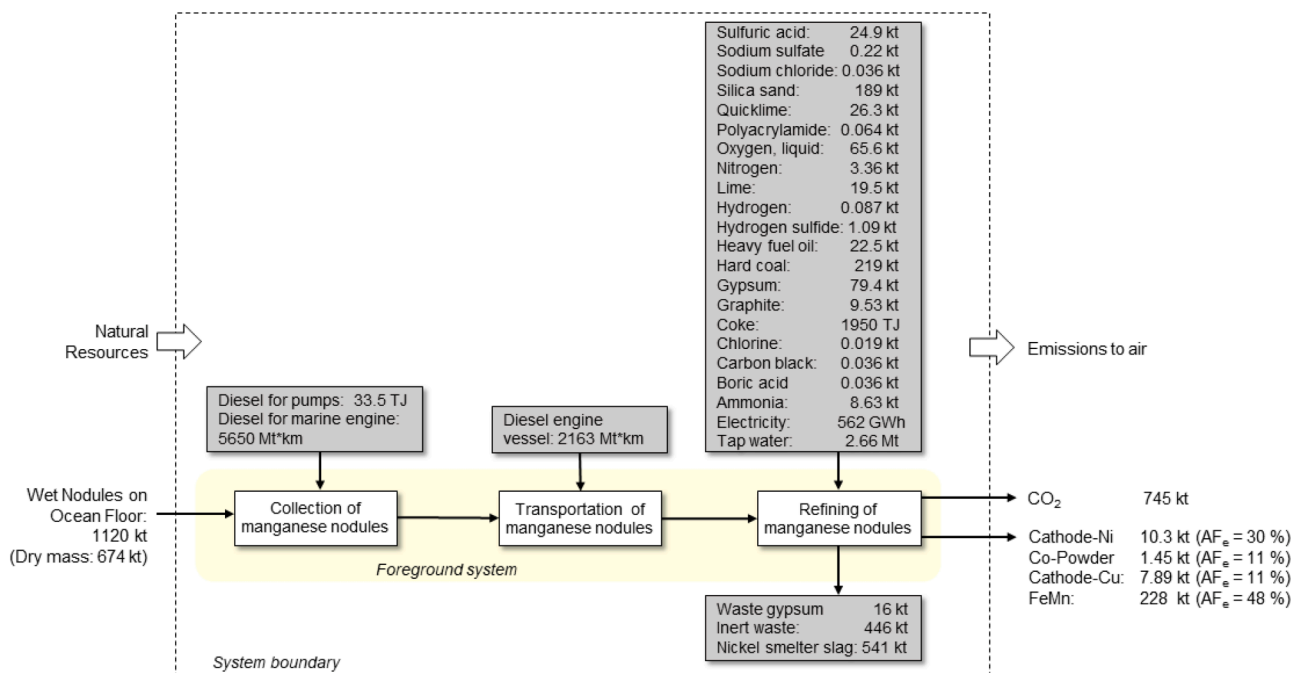


Fig. 1. Flow sheet of manganese nodules extraction scaled to annual production volume, from Dames & Moore and EIC Corporation (1977). Numbers based on Heinrich et al. (2020), Dames & Moore and EIC Corporation (1977), Kirchain et al. (2019).

2.2. Description of the comparative land-based product system

The system boundaries for the land-based projects were chosen in accordance with the manganese nodule model, i.e., cradle-to-gate. Metal production from manganese nodules represents a multi-product system whose product composition cannot be found in any land mine today. While the deep sea product system accounts for environmental impact for the entire production volumes of all the products (Ni, Cu, Co, Mn) from manganese nodules, the environmental impact of various land-based projects needed to be allocated and scaled to the equivalent product basket (for a detailed description, see supplementary material section III and Section 2.3). We included the respective mining and comminution stage in the foreground system, followed by individual paths for further processing, as displayed in Fig. 2. For the FeMn path and the Cu/Ni path, transport of intermediaries to refining plants were included. For the Ni/Co path, refining is performed at the mine site, so no transportation is needed.

The processes for providing various chemicals, energy sources and other auxiliary materials such as water, lime and graphite, as well as the treatment or disposal of waste are considered part of the background model. For the background systems, illustrated as grey boxes in Fig. 2, we used the ecoinvent v.3.7.1 cut-off database. As with the deep sea product system, capital goods and infrastructure were excluded from the scope unless the flows were listed in ecoinvent.

For setting up a comparative land-based product system, we modelled three existing mining projects based on the following criteria:

- Combination of metals in the mine's product mix include Ni, Cu, Co or FeMn.
- Process-specific data available.
- Mines are well-established and economically viable.
- Deposits and associated mining and production processes representative for the industry.

From this selection, the land-based product system was represented by a combination of the Kevitsa sulfide nickel project (Ni, Cu, and Co) in Finland, the Murrin Murrin nickel-cobalt project in Australia and the Woodie Woodie ferromanganese project in Australia. Further information on the mining projects and how they meet these criteria is provided in supplementary material section IV. The LCI for the land-based mining models and their underlying assumptions are thoroughly described in

supplementary material sections V to VII.

2.3. Allocation

Metals often occur as products of multi-output processes, as is the case for the product systems under consideration. As a result, the environmental impact of these processes must be attributed to all value-adding products. For LCA in metal production, the use of economic allocation is the predominant method to solve this issue (Tuusjärvi et al., 2012), because the ultimate objective of a base metal mine as a business is to generate monetary values. Thus, the procedure of economic allocation was applied throughout the foreground system of the land-based projects. We determined product prices based on average market prices of at least five years (supplementary material section VIII) in order to avoid an impact from price volatility. This time period can be regarded as sufficient, as Nuss and Eckelman (2014) have shown that allocation factors of most metals remain stable when comparing 5-year periods versus 15-year periods.

Since the declared unit is adapted to a product basket containing metal from 1 kg of dry manganese nodules, an assessment of the full product basket does not initially require allocation of the manganese nodule foreground system. However, to enable a comparison of the specific contributions of single metals, we supplemented our analysis by applying the same allocation procedure as for the land-based mines.

In addition, as part of the sensitivity analysis, we applied an alternative allocation for the manganese nodules module. All processes up to the point where the nickel-copper-cobalt bearing material flow separates from the ferromanganese bearing material flow (mainly pyrometallurgical reduction) were used for the alternative allocation scenario which we call "economic allocation, except FeMn reduction" (see supplementary material figure S 1 and Fig. 3). In detail, all on-site CO₂ emissions according to the in- and output tables from Dames & Moore and EIC Corporation (1977), less the specific CO₂ emissions of the neutralization with limestone (0.28 kg CO₂/kg limestone) were allocated to the material FeMn. The share of electricity for the production of FeMn was determined based on the sum of all rows in the process power column until FeMn production (see Dames & Moore (1996)). Coal, coke, silica, gypsum, oxygen and electrodes are further material inputs that were solely allocated to FeMn based on screening the flow diagrams in Dames & Moore (1996, section 9.1.2). This alternative was introduced as one iterative adjustment, since the hotspot analysis of the metal

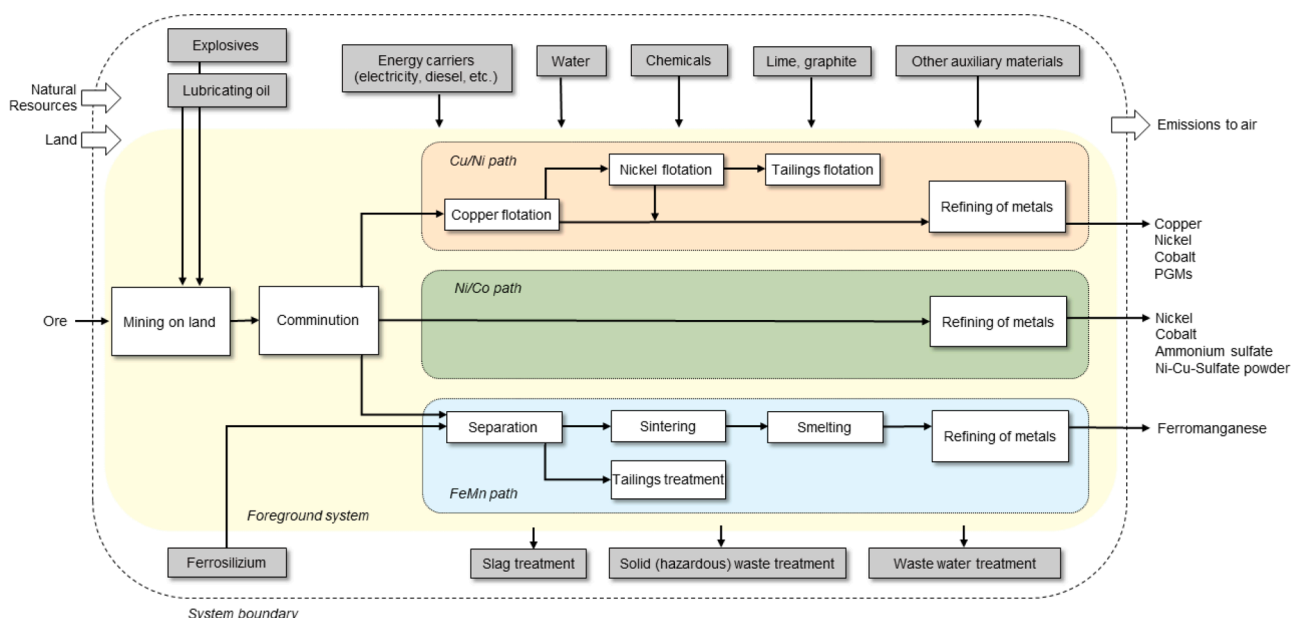


Fig. 2. Flow diagram of land-based mining and further processing of ores.

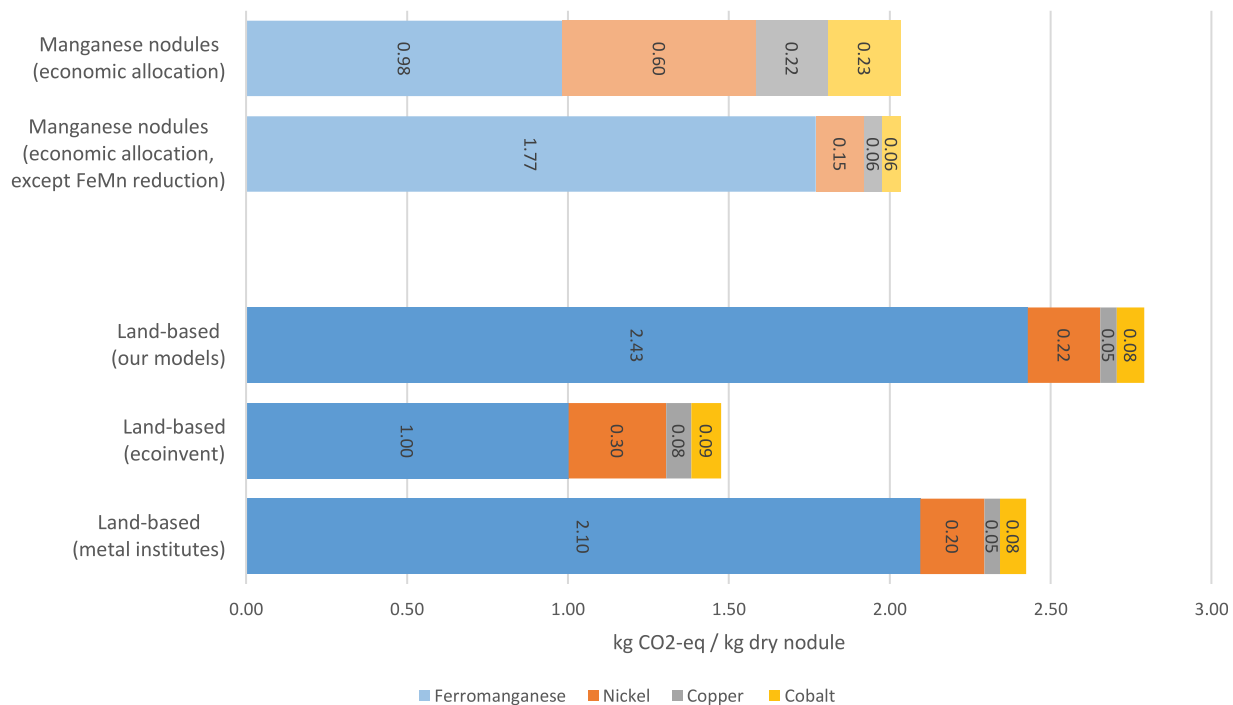


Fig. 3. Impact on climate change for the metals from the product basket for all investigated product systems in kg CO₂-eq/kg dry nodule.

production from manganese nodules (see Section 3.1) had shown that the CO₂ emissions from the coal-based reduction of FeMn cause more than half of the total climate change impact. Hence, we tried to unravel the flows and emissions specifically needed for the production of FeMn in the “black box” of metallurgical processing. Although this approach relies on knowledge in the field of, for example, process engineering or chemistry, there are no compelling reasons to choose one allocation method over another: neither economic allocation, nor our approach of separating FeMn can be seen as the better solution.

2.4. Comparison with data reported in the literature

To complement the comparison of the production of Ni, Co, Cu and FeMn from manganese nodules with their land-based counterparts, we conducted a database review on the latter's GHG impact. First, we collected generic data for each metal from the ecoinvent 3.7.1 cut-off database, which is widely used for modelling impact on climate change. Second, we complemented the data collection with data from metal-specific industry associations (Boonzaier, 2020; Cobalt Development Institute, 2016; International Copper Association, 2017; International Manganese Institute, 2014).

3. Results and discussion

The impact on climate change of metal production from manganese nodules is presented in Section 3.1, while the life cycle impact of land-based metal production is described in Section 3.2. Subsequently, we compare and interpret the results (3.3) under consideration of data quality (3.4), while the limitations of the study are pointed out in Section 3.5.

3.1. Impact on climate change of metal production from manganese nodules

According to our model, the total production of 367 g of product (total production volumes of Ni, Co, Cu, FeMn and Zn-Cu sulfide) from 1 kg of dry manganese nodules accounts for impact on climate change of 2.03 kg CO₂-equivalents (CO₂-eq). We define dry as 0% water content

here, also excluding chemically bound crystal water. At the given market price (five-year average) of the metals, we calculated the Ni-eq to be 0.051 kg per kg of dry manganese nodules. This leads to a climate impact of 39.1 kg CO₂-eq/kg Ni-eq.

Metallurgy is the process with the largest contribution at more than 95% (see last process in Fig. 1). A process hotspot analysis shows that CO₂ emissions from the reduction process of FeMn using coal contribute 55% to overall climate change impact. This process is comparable to a classic blast furnace steel-making process, which is known to emit enormous amounts of greenhouse gases. The electricity consumption during refining accounts for approximately one fourth of the total climate impact. This relatively high contribution of electricity consumption to the CO₂-eq is substantially caused by the high share of fossil energy use in the Mexican grid mix (>50%) (Ecoinvent 3.7.1, 2021). Further, the supply of oxygen accounts for roughly 5% of all emissions from metal production using manganese nodules.

To test the sensitivity of our model to the composition of the electricity mix, we calculated the LCA model for manganese nodules applying a grid mix with low climate impact. In order to have a realistic grid mix, we chose the Norwegian electricity mix, which has the lowest CO₂-eq (0.016 kg CO₂-eq/kWh) in ecoinvent although it is a mix of more than four different technologies. Recalculating all ecoinvent processes electricity providing in the foreground system to the Norwegian grid mix lowers the total climate impact by approximately 20%.

To obtain the CO₂-eq share of each metallic product in the product basket, we multiplied the production volume of each metal by its market price (five-year average) and calculated its total. The economically allocated results per kg of metal were scaled to the metal quantities in 1 kg of dry manganese nodule: namely, 15.3 g Ni, 11.7 g Cu, 2.15 g Co and 338 g FeMn. Using economic allocation, the climate impact contributes 0.98 kg CO₂-eq for FeMn, and 0.6, 0.22 and 0.23 kg CO₂-eq for Ni, Cu and Co to the overall impact, respectively. When using economic allocation for all metals except the FeMn reduction, the climate impact contribution of each metal changes to 1.77 kg CO₂-eq for FeMn, and 0.15, 0.06 and 0.06 kg CO₂-eq for Ni, Cu and Co, respectively. Hence, the environmental impact of FeMn, while those Ni, Cu and Co decrease accordingly.

3.2. Impact on climate change of land-based metal production

Based on our LCA models 2.78 CO₂-eq would be emitted if the equivalent amount of metals obtained from 1 kg of dry manganese nodules was produced from land-based production. The metal-specific climate impact is listed in Table 1.

The climate impact of Ni on average is 14.6 kg CO₂-eq/kg Ni including results from the Murrin Murrin project and the Kevitsa project. Co is also a product of these two projects where, according to our models, its production causes on average 39.3 kg CO₂-eq/kg Co (14.6 kg CO₂-eq/kg Ni-eq), slightly below the maximum numbers derived from the literature consulted. Cu production in Kevitsa results in 4.5 kg CO₂-eq/kg Cu (9.1 kg CO₂-eq/kg Ni-eq), which is relatively low in comparison to the range of figures from the literature but still exceeds the minimum CO₂-eq by the International Copper Association (2017). Compared to the other metallurgical processes, the Kevitsa project is characterized by low levels of on-site CO₂ emissions, because relatively few fossil fuels are used to generate energy. The electricity used is modeled as a low-emission fuel mix due to the high use of hydropower in Finland. Furthermore, sulfide deposits are generally associated with less energy use and GHG emissions, compared to lateritic deposits (Jessup and Mudd, 2008). As a result, Kevitsa's climate impact is mainly due to diesel fuel used for mining, with 45% of the overall CO₂-eq, followed by electricity with 27%.

In Murrin Murrin, 65% of the climate impact results immediately from the use of natural gas for electricity and industrial gas generation. Further, CO₂ is released in metallurgy from the use of lime for the neutralization of acidic process effluents. In addition, the supply chains of lime and sulfur also contribute indirectly but significantly to the climate impact (15%). Heat and electricity are generated on-site at Murrin Murrin by burning natural gas and elemental sulfur. Other relevant contributions to the CO₂-eq emitted stem from diesel combustion in ore mining (10%) and the use of electricity (also obtained on-site from steam turbines powered by natural gas) for crushing (6%). Since Ni and Co are produced by two models, Murrin and Kevitsa, and different climate impacts were obtained in each case, we used the arithmetic mean to represent the specific climate impact per metal. The exploitation of lateritic (Murrin-Murrin) versus sulfidic (Kevitsa) deposits is about 60/40 (Butt and Cluzel, 2013).

The production of FeMn based on data from the Woodie Woodie project emits 7.19 kg CO₂-eq/kg FeMn (96.2 kg CO₂-eq/kg Ni-eq). This result exceeds the maximum result from the literature sources provided by the International Manganese Institute (2014). The pyrometallurgical processing used for the production of FeMn is the key contributor to climate change. More than 75% of the CO₂-eq is attributable to the use of Chinese electricity, whose mix is dominated by fossil fuels, and the use of coal and coke for the reduction of oxides in this process. The modeled heat generation is primarily based on oil and coal.

Again, we performed a sensitivity analysis for all the land-based LCA models, applying the Norwegian grid mix as a low climate impact electricity scenario (see Section 3.1). This decreases the CO₂-eq for Woodie Woodie by 50% and for Kevitsa by 25%. As Murrin Murrin is a full off-grid operation in which sulfur and natural gas are used for the energy demand, but also to produce other industrial gasses, it is very complex to alter the electricity supply to the Norwegian grid mix.

Although this might not be optimal for analyzing an alternative electricity scenario, it illustrates very well the reality of the metal industry as it shows how diverse metal production projects are. In order to check the magnitude of this error, we tested an extreme scenario in which the energy equivalent of all the natural gas used in Murrin Murrin is changed to simulate the Norwegian grid mix. Our analysis showed that the error related to the total impact on climate change for the product basket is less than 5%.

Although the system boundaries in all of the comparative datasets are the same as our models – namely, cradle-to-gate including mining, beneficiation and refining – the results vary. Because FeMn contributes by far the most to the overall climate impact, we want to briefly describe some possible reasons for the differences in the CO₂-eq/kg FeMn. Theecoinvent dataset assumes a 75% utilization of electric arc furnaces for the ferromanganese production. Additionally, the energy for the electric arcs is provided by 66% hydropower. Therefore the use of fossil energy carriers as well as reductants is low compared to our models and those of the International Manganese Institute (2014). The International Manganese Institute (2014) datasets are based on data surveys from 17 different mines and smelters in different countries. The LCA background data in the study's results is a mix of GaBi (2014), ecoinvent as well as Australian and United States governmental LCA data. Even though the metallurgical processing of our own FeMn model is mainly based on the LCI of the International Manganese Institute (2014) study, the sole fact that we use ecoinvent as background data and that we adapted the datasets to the refinery location in China explain the differences in the results.

3.3. Climate impact comparison of metal production from land mines and from manganese nodules

Fig. 3 shows the climate impact for producing an equivalent product basket: from land mines based on our models (middle bar), from average figures from the literature (two bars at the bottom) and from manganese nodules with both alternative allocation approaches (two bars at the top).

Contrasting the overall climate impacts from manganese nodules (bar at the top) with the three land-based alternatives, we obtain opposite interpretations for the comparisons. On the one hand, the overall climate impact from manganese nodule production is 16% lower when compared with the results for land-based metal production (bar at the bottom) published by the metal institutes, or approximately 27% lower when compared to land-based metal production from our model (third bar). On the other hand, when the metal production from manganese nodules (bar at the top) is compared to the land-based ecoinvent 3.7.1 results (fourth bar), the CO₂-eq of the nodule production are 28% higher.

From this analysis, we conclude that the production of Ni, Co, Cu and FeMn from manganese nodules could create either a higher or a lower impact on climate change compared to land-based deposits. The difference between the product systems, and hence their specific assumptions and background data is, however, not significant (approximately 25%) given total uncertainty associated with LCA. In general, LCA results are scattered as a consequence of freedom in methodological choices, background data, software and study design (Sanjuan-Delmás

Table 1

Impact of metals on climate change, based on our models and other literature. Note that the average is the arithmetic mean.

Metal:	Ni	Cobalt	Copper	Ferromanganese
Unit	kg CO ₂ -eq/kg Ni	kg CO ₂ -eq/kg Co	kg CO ₂ -eq/kg Cu	kg CO ₂ -eq/kg FeMn
Our models	Kevitsa: 9.15 Murrin: 20.1 Mean: 14.6	Kevitsa: 24.6 Murrin: 53.9 Mean: 39.3	Kevitsa: 4.46	Woodie: 7.19
ecoinvent 3.7.1 cut-off	20	39	7	3
Industry Association ¹	13	38	4.1	6.2
¹ references	(Boonzaier, 2020)	(Cobalt Development Institute, 2016)	(International Copper Association, 2017)	(International Manganese Institute, 2014)

et al., 2022; Speck et al., 2015; Takano et al., 2014). Using copper as an example, Sanjuan-Delmás et al. (2022) have shown that different combinations of software and databases lead to considerable variations in results (~ 26% difference for the impact category of climate change). Additionally, the climate impact of metal production includes uncertainty arising from a variety of different technologies, countries, energy carriers and deposits, to name just a few sources (Skarn Associates Limited, 2020; World Gold Council, 2018).

Similar to the exploitation of manganese nodules (see Section 3.1), for land-based production of Ni, Cu, Co and FeMn, metallurgical processing is the major contributor to climate change (70–85%) (Boonzaier, 2020; Cobalt Development Institute, 2016; Ecoinvent 3.7.1, 2021, market for copper cathode, International Manganese Institute, 2014). As the basic metallurgical processing of manganese nodules is similar to that of existing metal smelters (e.g. treatment of Ni laterites, see Sharma (2017)), climate impact is more dependent on the properties of metallurgical processing (e.g., sources of energy used, process efficiency or pyro- vs. hydro route) than the origin of the ore (deep sea or land). Therefore, lower climate change impact depends on the metallurgical processing using state-of-the-art technology, like high electrification with regenerative power sources and H_2 as a reducing agent instead of fossil sources. A completely different processing route based on hydrometallurgy could also lead to lower climate change impact.

A comparison of the alternative electricity scenarios applied to both product systems shows that the climate impact approaches the dry nodule value of 1.5 kg CO_2 -eq/pro kg. Consequently, the differences between the results of our LCA model for manganese nodules and for land-based mining are minimal (for a more detailed graph showing results for the alternative electricity scenario, see supplementary material figure S 3). Based on our assumption that the transition to low-carbon electricity mixes will apply to both product systems in the future, we conclude that the climate impact of both manganese nodule production and land-based production could decrease. This means that any efforts to push forward the transition to low-carbon energy sources in the electricity sector will lead to a similarly lower climate impact of metal production from land-based ores compared to pristine manganese nodules.

Application of the two allocation types to the manganese nodule model in Fig. 3 shows that allocation strongly influences the results of comparing individual metals (different proportions of the metals). Taking this into account when comparing the individual metal results from both allocation types with the land-based results, no general statement can be made about the specific climate impact of each individual metal being lower or higher than the production from land-based sources. E.g., copper from the manganese nodules mining with economic allocation (Fig. 3, first bar) has a higher climate impact than its counterparts in all three land-based models (Fig. 3, lowest three bars). But, the climate impacts of copper from manganese nodules mining with the alternative allocation method (Fig. 3, second bar) is slightly lower than its land-based counterpart based on ecoinvent (Fig. 3, fourth bar). As all mining projects discussed in this study without for Woodie Woodie, are multi-output systems, the allocation approach chosen is highly relevant for their results. Accordingly, caution should be taken when comparing the climate impact of individual metals from different sources. Although FeMn is the main product (by mass and total monetary value), CO_2 -eq in the economic allocation model is relatively low in comparison to results obtained under application of the alternative allocation scenario “economic allocation, except FeMn reduction”. FeMn has the lowest market price of the given metal products. Further, supplying considerable amounts of FeMn from deep sea mining to the market might affect market prices. A decrease in FeMn market prices would result in further lowering the respective allocated CO_2 -eq (economic allocation). One should note that the composition of metals shown in the third bar (our models) would also change when applying a similar allocation approach to Kevitsa and Murrin, as they also represent multi-product projects.

Furthermore, we can see in Fig. 3 that the production of FeMn plays a

dominant role, accounting for the highest share of the impact in each production pathway but specifically for the land-based metal production using our models and for the manganese nodules with the alternative economic allocation scenario without FeMn reduction (87%, second and third bar). This high share derives from the large amount of FeMn produced from manganese nodules when following a zero-waste approach with a pyro-route for FeMn. The large amount of manganese leads to a high monetary value share in the nodule which is, despite the lower market price, nearly equivalent to the sum of the total monetary value of Ni, Co and Cu. Consequently, an equivalent ratio of metals has to be produced from land mines to obtain a comparable product basket. The high climate impact we calculated for FeMn production based on the model for Woodie Woodie is thus one factor that explains the approximately 27% higher CO_2 -eq of the land-based model in comparison to the manganese nodule model. Fig. 4 supports this observation by comparing the climate impact of each mining project in relation to the total monetary value of its products, which is represented by the amount of Ni-eq produced.

The model of FeMn production (Woodie Woodie) has the highest impact on climate change, with a CO_2 -eq per Ni-eq at least twice as high as the other models. This result is in line with the results shown in Fig. 3, in which FeMn from land mines was also the only metal to have higher climate impact than its deep sea counterpart (top bar, economic allocation). The CO_2 -eq per kg Ni-eq in Kevitsa and Murrin Murrin is lower (at least half as much) in comparison to the deep sea mining results, because nodule processing includes the pyrometallurgical processing of FeMn, which is highly CO_2 -intense.

3.4. Data quality assessment

Due to the prospective character of the technologies for mining and processing manganese nodules, we used literature and feasibility studies closely representing the technologies chosen as a data source for the inventory. Although the studies have not been carried out recently, a critical review by industry experts in this study assessed the inventories as reliable and complete. However, regarding the geographical correlation of the inventory with future production, major uncertainties remain until real production capacities have been established. The literature used is mainly based on techno-economic analysis of mining projects and were not specifically collected with the purpose of doing a study on climate impact. However, the probability that relevant sources of greenhouse gases are missing is low.

In reviewing the carbon balance for deep sea mining metallurgy reported by Dames & Moore and EIC Corporation (1977), we were not able to fully reproduce their reported emissions from carbon sources. Laboratory tests with manganese nodules show that the metal FeMn contains a maximum of about 7% carbon. This carbon is firmly bound in the crystal structure of the metals and thus offers no potential for further CO_2 emissions into the atmosphere. However, this does not yet explain the entire difference. We could not definitively clarify whether some of the carbon remains in a solid state or if there are additional emissions that should be accounted for. We countered this uncertainty by applying the lower and hence more optimistic figure as reported by Dames & Moore (1996), concluding that our results potentially underestimate the climate impact of the metal production from manganese nodules by 15%.

To assess uncertainties introduced from the land-based LCIs, we performed a data quality assessment for the three mining projects based on the Pedigree matrix (Ciroth et al., 2016). The procedure and the results are thoroughly described in supplementary material section X. Despite the relatively high GWP result for our FeMn model, which significantly affects our results, the good data quality of the Woodie model increases confidence in the reliability of the result.

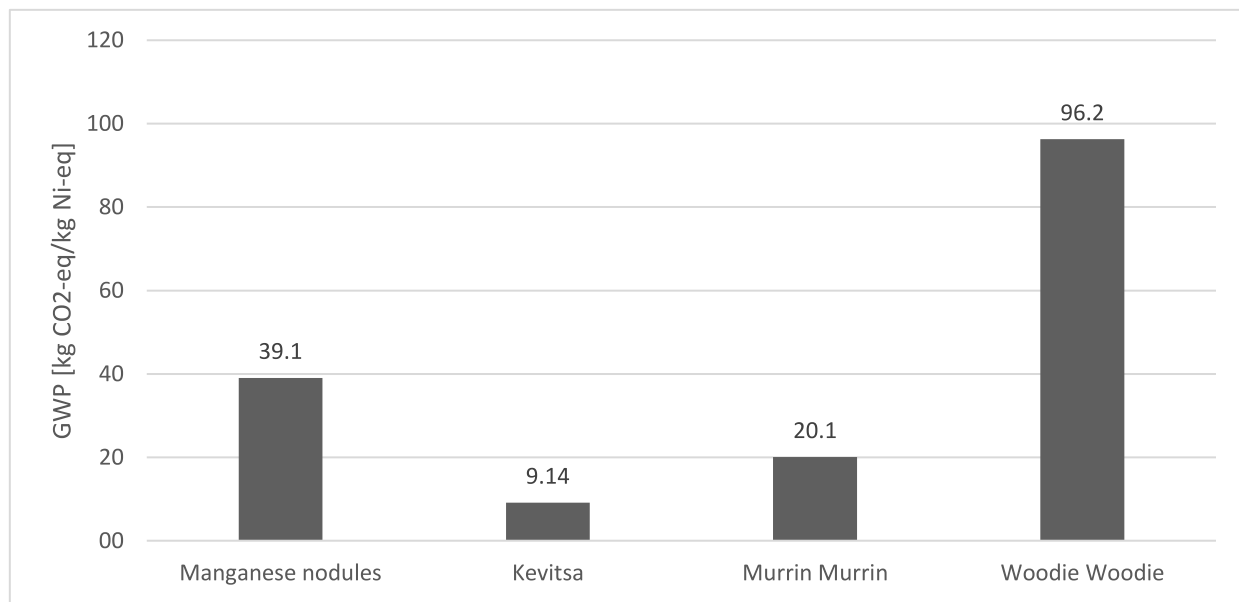


Fig. 4. Impact on climate change of the three land-based projects and the manganese nodules project per kilogram of nickel-equivalent (Ni-eq).

3.5. Limitations

In exploiting manganese nodules, it is crucial to protect the world's oceans, in particular their biodiversity, although this was not considered here. According to [Pr  at et al., \(2021\)](#), the impact on biodiversity of such deep sea mining cannot be reliably mapped with the LCA instrument at present. We propose an initial method for integrating biodiversity in deep sea systems into LCA in the future. However, other environmental impacts that were not part of the scope of this study should also be studied, as otherwise trade-offs between impact categories might be overlooked. For example, like many other mines, the Murrin Murrin project uses large amounts of sulfur to provide energy and sulfuric acid. Energy recovery from sulfur does not produce CO₂-eq, which is positive from a climate perspective, but this procedure increases the environmental impact on acidification potential (AP, in SO₂-eq). We also did not investigate impact by plumes, or noise and light, which are slow processes with long time spans that are often discussed in the field of manganese nodule mining. The same applies to land-based metal production, for which other impacts like freshwater, toxic wastes, land use change and social issues have been observed, too.

In general, comparing the exploitation of manganese nodules with land-based production poses difficulties, as the metallurgical ore of manganese nodules is very different from terrestrial deposits ([Sharma, 2017](#)). Therefore, we used an approach comparing the exploitation of manganese nodules with specific land-based mining projects with production volumes and metal composition as similar as possible to manganese nodules, which allowed us to use comprehensive and reliable data instead of using generic LCA datasets. Another way to compare the impact on climate change from the production of metals from manganese nodules versus land-based deposits would be to compare it with common CO₂-eq from LCA databases and studies. However, one must accept that the often generic data sets are based on different assumptions, extrapolations (for intersystemic data scaling, see [Fritz and Schmidt \(2021\)](#)) and parameters (such as ore grade, stripping ratio and degree of electrification). In addition, although we crosschecked our results with CO₂-eq from other studies as well as the databases [Ecoinvent 3.7.1 \(2021\)](#) and [Skarn Associates Limited \(2020\)](#), we did not include prospective aspects or scenarios, e.g., market changes or different countries for production sites. This choice better fits the purpose of our study to compare the exploitation of manganese nodules to land-based projects.

4. Conclusion

This study sets out to analyze the impact on climate change by the production of Ni, Cu, Co and FeMn from manganese nodules from the Clarion Clipperton Zone (CCZ) compared to land-based production.

Our study identified that the climate impacts from the production of Ni, Cu, Co and FeMn from manganese nodules is 16% lower than the climate impacts from land-based metal production with data published by the metal institutes, or approximately 27% lower when compared to land-based metal production with data from our model. In contrast, when the metal production from manganese nodules is compared to the land-based results of ecoinvent 3.7.1, the CO₂-eq of the nodule production is 28% higher. Therefore, due to these variations in climate impact for metal production, decision-making regarding future manganese nodule exploitation should not be based on climate change. From a climate change perspective, the production of metals from manganese nodules – depending on the process routes as well as the specific land-based mine chosen for comparison – has a divergence of 25% both ways. Such scattered LCA results are simply not good enough to be a basis for decision making.

Our research has also shown that when following the recommended processing route, FeMn is the highest metal contributing to climate impact in all the production scenarios analyzed. The processing of FeMn is therefore highly relevant for future discussion about climate impact and whether alternative sourcing from the deep sea is a potential substitute for or could complement land-based production.

A further study could profitably assess the impact of the metallurgical processing of manganese nodules on climate change by using a metallurgical process simulation tool like HSC Sim ([Outotec Research Center, 2020](#)). These tools can implement LCA databases (including CO₂-eq) in the modelling process. It would be possible to gain in-depth insights on different smelting and refining routes and their hotspots in terms of climate change impact. The implications of the many uncertainties and assumptions in the comparison of the production of metals from manganese nodules versus land-based ores could be further improved by additional studies with different scenarios. Several questions remain unanswered regarding market prices and their implications for the results through, e.g., economic allocation. For example, the results of this study could be combined with market forecasts for the respective metals. Here, one could also consider how the interrelationship of Ni, Cu, Co and FeMn market prices could potentially change as a

result of future manganese nodule mining.

Based on our study, we conclude that climate change should not serve as a pivotal argument either for or against the realization of real manganese nodule projects. Most importantly, in comparison with land-based mining, metal production from manganese nodules does not decrease impact on climate change at a significant level that would justify other environmental impacts and risks.

CRedit authorship contribution statement

Benjamin Fritz: Methodology, Software, Validation, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Pia Heidak:** Validation, Writing – original draft, Writing – review & editing, Visualization. **Jürgen Vasters:** Validation, Formal analysis, Investigation, Writing – review & editing. **Thomas Kuhn:** Validation, Writing – review & editing. **Gudrun Franken:** Validation, Resources, Writing – review & editing. **Mario Schmidt:** Conceptualization, Methodology, Validation, Funding acquisition, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

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